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Environmental factors influencing annual sucker (*Catostomus* sp.) migration into a Great Lakes tributary

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ABSTRACT

Fish migration in rivers is a growing area of concern as mounting anthropogenic influences, particularly fragmentation from dams and barriers, constitute major threats to global river species diversity. Barriers can impede the movement of fishes between areas critical to the completion of their lifecycle, affecting both population and ecosystem viability. In response, fish passage solutions have been identified as a critical need to maintain fisheries viability in the Laurentian Great Lakes, and around the world. Pivotal to the success of these fish passage solutions is a more complete understanding of the movement phenology and environmental cues that instigate migration. We used a dual-frequency identification sonar (DIDSON) to evaluate environmental triggers of river entry during spring and summer for three size classes of migratory fishes in the Boardman River, a Lake Michigan tributary. Our results indicate that medium size fish (>30 cm and < 50 cm), primarily composed of white sucker *Catostomus commersonii* and longnose sucker *Catostomus catostomus* were 21% more likely to enter the river at sunset and 25% less likely at midnight in comparison to midday. Entry rates of medium fish increased 6% for every 1 °C increase in river temperature, 4% for every 1 m^3/s increase in river discharge from the day prior, and were reduced by 1% for every 10 cm increase in lake level. Understanding these processes in the tributaries of the Great Lakes is important to inform the fish passage solutions currently being developed for the Boardman River, and to inform management regulations for Great Lakes migratory fishes.

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Introduction

Migration is a term often applied to adaptive, long distance movements that occur predictably in the life cycle of a species, but can become increasingly difficult to define due to the panoply of disciplines that use it and its exhibition across numerous animal taxa (Baker, 1978; Dingle, 1996; Smith, 2012). At its most basic level migration is simply the act of moving from one spatial unit to another (Baker, 1978). As such, migration is a common phenomenon present in numerous fish species exhibiting a variety of life history strategies. Although the specific purpose can differ, fish migration is often linked to reproduction (Dingle, 1996; Sinclair, 1983). Fish migration within riverine environments is a growing area of concern as mounting anthropogenic influences constitute a major threat to global river species diversity (Vörösmarty et al., 2010). In particular, the fragmentation of river systems alter hydrologic conditions (Kemp and O'Hanley, 2010) and can impede the movement of fish between areas used for feeding and spawning (Liermann et al., 2012). In response, fish passage solutions have been identified as a critical need to maintain viable fisheries around the world. Pivotal to the success of these fish passage solutions is a more complete understanding of the movement phenology and environmental cues that instigate migration.

River entry of migratory fishes is influenced by multiple environmental variables including water temperature, river discharge, and photoperiod (Jonsson, 1991; Legget, 1977), and that influence can vary among species because of varying physiologies and life history strategies (Lucas et al., 2001). Typically, increasing and decreasing photoperiod is the predictive, proximate factor indicating to the fish the season of migration (Eriksson et al., 1982). Photoperiod is the same on each specific day every year; however, inter-annual variation in fish migration timing is often observed to varying extent highlighting the importance of understanding the influence of other environmental variables. Many species of fish in the Laurentian Great Lakes (hereafter Great Lakes) migrate into rivers during the spring and the timing of river entry is influenced by the dynamic environmental conditions present this time

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of year. For example, water temperature seems to be the primary trigger to upstream movement for migratory rainbow trout *Oncorhynchus mykiss* (Workman et al., 2002) and sea lamprey *Petromyzon marinus* (Binder et al., 2010) during the spring in the Great Lakes. However, the temperature threshold for migration has been shown to vary between species and water level has also been documented as predictive in some systems for sea lamprey (Binder et al., 2010; Workman et al., 2002). It has also been well documented that white sucker *Catostomus commersonii* and longnose sucker *Catostomus Catostomus* migrate into Great Lakes tributaries in the spring (Burtner et al., 2011; Childress et al., 2014, 2016; Childress and McIntyre, 2015), and studies across their range have shown that these migrations require a minimum threshold temperature and adequate water levels (Barton, 1980; Corbett and Powles, 1983; Geen et al., 1966; Raney and Webster, 1942). However, to our knowledge no studies of the environmental drivers of sucker migrations exist for Great Lakes tributaries. Although much is known about upstream migration of some species, further understanding of the environmental cues for fish migration into Great Lakes tributaries at higher spatial and temporal resolutions is needed (Landsman et al., 2011), especially in and around fishways.

Enhanced understanding of the environmental cues that trigger fish movement in Great Lakes tributaries is important because it can improve fisheries management practices intended to support recreational, commercial, and subsistence fisheries, native fish restoration, and invasive species control. For example, fisheries managers need to know when and at which life stage fish are moving into the rivers to establish closed seasons and catch limits. Additionally, knowing when native or invasive species move upstream can inform the operation of selective fishways and hydropower facilities that protect native or economically valuable fishes (e.g., lake sturgeon *Acipenser fulvescens*, walleye *Sander vitreus*) while blocking invasive species (e.g., sea lamprey) (Vélez-Espino et al., 2011). Integrated pest management (IPM) has been utilized across the Great Lakes to control sea lamprey and has reduced their numbers to 10% of historic peaks (Siefkes, 2017). However, resource managers across the Great Lakes basin often consider alternatives to the use of both lampricide and low-head barriers which have proven to be two of the most pivotal tools to achieve current reductions in sea lamprey populations (Siefkes, 2017; Zielinski et al., 2019). To date, alternative barriers (i.e., seasonal barriers, velocity barriers, electric barriers) have been intermittently successful in the sea lamprey control program (McLaughlin et al., 2007; Zielinski et al., 2019). The ability to differentiate movement phenology and its environmental drivers between invasive and native species lends itself to IPM of sea lamprey and other invasive species through the further development of selective passage.

Located in northern Michigan, the Boardman River is an important system for improving knowledge of fish migration because of its fish productivity potential and value to the local community. Because of its value to both recreational and tribal stakeholders, this river is the focus of the Boardman River Ecosystem Restoration Project (https://www.theboardman.org), aimed at improving habitat connectivity by removing three dams and modifying the lowermost barrier, the Union Street Dam. Union Street Dam will be the future site of a one-of-a-kind research facility called FishPass that aims to develop solutions for bi-directional selective fish passage through controlled sorting (https://www.glfc.org/fishpass.php). The end result of FishPass is to restore connectivity for migratory and non-migratory movements of desirable fish (e.g., species native to the Great Lakes) while blocking undesirable fish (e.g., invasive sea lamprey). Understanding movement phenology of fish in the Boardman River and its drivers provides a starting point from which controlled sorting may begin at FishPass.

Populations of numerous fish species could benefit from increased habitat connectivity in the Boardman River, which connects a recently restored tributary with a large bay of Lake Michigan. In addition to benefiting fish populations, increasing connectivity has the potential to benefit the aquatic ecosystem in the Boardman River as a whole through increased movement of nutrients upstream via migratory fish vectors. Migrations of iteroparous suckers (white sucker and longnose sucker) act as an upstream vector of nutrients in the Great Lakes which increases algal and invertebrate growth (Childress et al., 2014); these nutrient subsidies are a result of excretion and gamete deposition during the spring reproductive migration (Childress and McIntyre, 2015). While anecdotal accounts have documented large numbers of suckers present in the spring below Union Street Dam, little is known about sucker migration in the Boardman River. If large numbers of suckers migrate into the Boardman River similar to the hundreds to tens of thousands documented migrating in other tributaries of Lake Michigan (Childress et al., 2014, 2016; Childress and McIntyre, 2015), as the data presented here suggest, then understanding their movement phenology and its drivers is critical to the success of selective fish passage (i.e., FishPass) and ecosystem restoration.

Dual-frequency Identification Sonar (DIDSON) has been used to observe and understand fish behavior such as spawning, feeding (Becker et al., 2013), and migration (Baumgartner et al., 2006; Martignac et al., 2015; McCann et al., 2018). The goal of this study was to use DIDSON to evaluate environmental triggers of river entry for three size classes of migratory fishes in the Boardman River, a Lake Michigan tributary. Multiple size classes were enumerated separately to evaluate triggers on different species of migratory fish because species recognition was not possible. We hypothesize that river entry timing of fish is influenced by photoperiod, discharge, temperature, and river stage. Consistent with the idea that environmental factors influence river entry differently depending on species, we predict that the three size classes of fish will be affected by environmental covariates differently. Evaluation of fine-scale effects on the timing and covariates of fish entry in the Boardman River provides vital information on the natural phenology of fish movement that can help to inform FishPass operations and fisheries management throughout the Great Lakes.

Methods

Study design

A DIDSON was deployed to record the number and timing of fishes entering the Boardman River. Because species recognition was not possible, fish were enumerated by three size classes: small ($\leq$30 cm), medium (>30 cm and < 50 cm), and large ($\geq$50 cm). These size classes were selected with the primary intention of using the medium size class to represent the anecdotally documented large spring migration of suckers. Electrofishing surveys were used to understand the fish community composition and determine what species are present within each size class. Passive integrated transponder (PIT) tag detection data were used to compare the detection timing of individual species in comparison to DIDSON observations. The electrofishing and PIT tag data were used in combination to further elucidate the species composition within size classes and make inferences about the fish migrations being captured by the DIDSON. Environmental data were collected to characterize the conditions fish experienced during stream entry. These data were used as covariates in a hazard analysis to determine the environmental cues for river entry of fish in a Great Lakes tributary.

Study site

A tributary to Lake Michigan's Grand Traverse Bay, the Boardman River is located in Grand Traverse County of Michigan's Lower Peninsula, USA. The river is approximately 45.4 km long, with a watershed area of approximately 743 km^2. Data were collected approximately 1.25 km upstream of the mouth of the Boardman River at the James P. Price Trap, Transport and Harvest Facility (44.764505° N, 85.627031° W; Fig. 1). We assume fish observations at this location are generally representative of recent entry to the stream given its intermediate proximity between the mouth (1.25 km) and the dam (0.6 km), although complexity of fish movement (e.g., bi-directional movements, milling behavior, etc.) may affect this assumption.

DIDSON

A DIDSON (Standard Version 300 m; Sound Metrics Corp., Bellevue, Washington, USA) was deployed in high-frequency mode (1.8 MHz) in the spring and summer of 2017–2019. To reduce temporal bias, analysis was restricted to Julian Days 79–181 (March 20 – June 30) in all years. A sampling window of 10 m was used to ensure high image quality to identify fish size. The DIDSON was mounted in the thalweg approximately halfway down the water column and the field of view was positioned horizontally and perpendicular to stream flow at a downward tilt of −4.7° with a viewing window spanning from 1.67 m to 11.67 m (Fig. 1). The resulting field of view covered approximately 33% of the 33.9 m^2 cross-sectional area of the stream. The bottom channel morphology within the viewing window was flat with a substrate containing mostly sand and small boulders.

PIT tag telemetry

A PIT antenna array was installed in August 2018 using a Multi-Antenna half-duplex reader (Oregon RFID, Portland, Oregon, USA) and four rectangular pass-through antennas (4–6 m L × 1.25 m W) which spanned the entire width of the river channel approximately 2 m downstream of the DIDSON unit (Fig. 1). Each antenna consisted of 2 loops of 12-gauge stranded wire tuned to an inductance of ~ 60–80 μH. All antennas were connected to tuning modules and connected to the PIT tag reader by twinaxial cable. Each time a tagged fish passed through an antenna, the time of passage, PIT tag identification number, antenna number, and time between detections were logged onto a memory card.

Environmental monitoring

Stream temperature, river stage, and river discharge were measured to determine their relative influence on entry rate of fishes in the Boardman River. Continuous stream temperature (to the nearest °C) was collected hourly at the DIDSON site using a multi-parameter water quality SONDE (model: 6600 V2-4; YSI Inc./Xylem Inc., Yellow Springs, Ohio, USA). River stage data (cm above sea level) were collected from a HOBO data logger (model: MX2001-04; Onset Computer Corp., Bourne, Massachusetts, USA) at the river mouth, approximately 1.06 km downstream of the DIDSON site. River stage data were rounded to the nearest 10 cm above sea level to reduce observed variation associated with the influence of wave action. Discharge (m^3/s) at the Union Street Dam was determined from a direct drainage area ratio adjustment from a U.S. Geological Survey stream gauge (04127200), located at Bitner Road in Traverse City, Michigan. Both discharge and river stage were used in our analyses because discharge was a good representation of river conditions whereas river stage was not only influenced by discharge, it was also influenced by overall water levels in Lake Michigan. Therefore, river stage was used as a proxy for the conditions fish experienced in the lake prior to river entry.

Physical sampling of fishes

Six electrofishing surveys were conducted in the Boardman River below Union Street Dam within the DIDSON sampling period (March 20 - June 30). These surveys were used to determine the

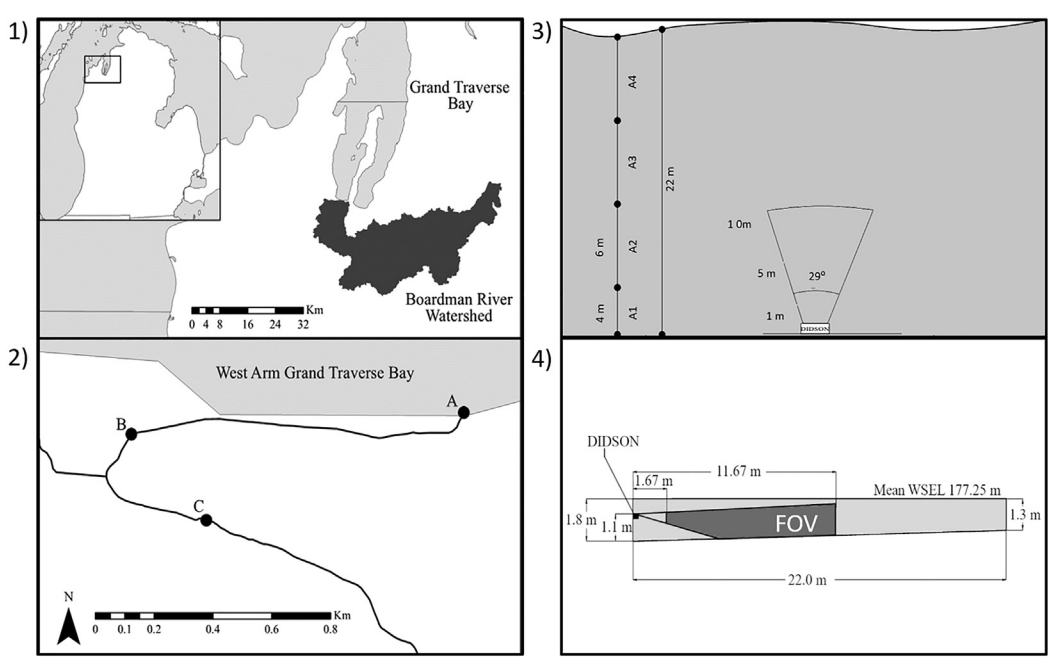

Fig. 1. Panel 1: Map of the Boardman River watershed; Panel 2: Map of the Boardman River study site showing A) the Boardman River mouth, B) the location of the DIDSON (44.764505° N, 85.627031° W), and C) the location of the Union Street Dam; Panel 3: General diagram (aerial view) of the DIDSON and passive integrated transponder (PIT) antenna array setup; Panel 4 General diagram (cross section) of the DIDSON field of view (FOV) under mean water surface elevation (WSEL) conditions.

species composition of fish enumerated during DIDSON analysis. Surveys occurred in April 2019 (n = 2), May 2017 (n = 1) and 2019 (n = 2), and June 2019 (n = 1). Surveys followed the Michigan Department of Natural Resources (MIDNR) electrofishing sampling protocol for random sampling sites in non-wadeable rivers (Wills et al., 2011). Each survey was conducted during daylight hours and consisted of a single pass with an electrofishing boat and fish collection using 4.77 mm delta mesh dip nets. The length and species identification of all remaining fish were subsequently recorded and fish were released. In some instances, only the first 50 individuals of each species were measured (Table 1). Fish captured over 150 mm, from July 2018- June 2019 were implanted with individually coded 23 mm PIT tags (Oregon RFID, Portland, Oregon), and released until a sample size of approximately 50 individuals was reached (Table 2). Chinook *Oncorhynchus tshawytscha* and coho salmon *O. kisutch* were not PIT tagged due to semelparous spawning migration, and are removed by the MIDNR in the fall during their fall spawning migration downstream of the DIDSON. Protocols involving the handling of fishes were carried out in accordance with United States federal guidelines for care and use of animals and were approved by the American Fisheries Society (Jenkins et al., 2014).

DIDSON data processing

The DIDSON data were manually examined to quantify fishes as they swam through the field of view using the DIDSON V5.25 software. Fish on the cusp of the different size classes were measured digitally by drawing a line along the body with the mark fish tool in the DIDSON V5.25 software and assigned accordingly. Digital estimates of fish lengths derived from DIDSON have been found to be similar to true length using similar settings to those used here (field of view < 15 and 1.8 MHz; Burwen et al., 2010; Daroux et al., 2019; Grote et al., 2014; Han et al., 2009; Hightower et al., 2013). All fish moving in the DIDSON field of view, regardless of direction, were counted. Images were first evaluated at 16 frames per second but observers were allowed to increase or decrease playback speed as necessary to identify individual fish. Furthermore, reviewers were allowed to re-watch clips as many times as necessary until they were confident in the number of fishes obtained in all size classes. Due to the complex variability of swimming behaviors observed (i.e., schooling, milling, bi-directional lateral and longitudinal movement), direction of travel was not noted during manual processing, but most movement was upstream. Fish exhibiting undirected swimming throughout the field of view were enumerated once unless they left the field of view. If a fish left the field of view and re-entered, the observer could not verify its individual identity, and thus it was counted as another discrete event. It is important to note that milling behavior has the potential to reduce overall accuracy of enumeration (Magowan et al., 2012; Pipal et al., 2012).

Over the course of the sampling period, the DIDSON recorded>1700 h of data per year. Manual inspection of the entire data set was not feasible, so the data were subsampled into four, two-minute DIDSON clips per day. As a result, approximately 1% of the sampling period was subsampled. These two-minute clips were systematically selected at sunrise, sunset, the midpoint between sunrise and sunset (i.e., mid-day), and the midpoint between sunset and sunrise (i.e., mid-night). A recent study by Eggleston et al. (2020) demonstrated that extrapolating the average fish count from three two-minute sub-samples was not statistically different than manually counting the full hour subsampled. Taken in combination with the result of another evaluation of DIDSON by Petreman et al. (2014), which found that systematic seasonal sampling resulted in a small increase in accuracy compared to a random subsampling at low sampling efforts, supports the sampling scheme utilized in this study. Inspection of the entire data set would be preferred in the absence of temporal and financial constraints. All clips were randomized prior to manual inspection and inspected by an independent observer each year trained using the same data set to reduce potential bias if an observer's ability to quantify fish changed through time. Large inter-observer bias has previously been documented in some instances (e.g., Keefer et al., 2017) but minimal observer bias has been reported in others, indicating that multiple individuals can be used interchangeably for manual processing of DIDSON data (Cronkite et al., 2006; Petreman et al., 2014). To reduce potential bias introduced from using multiple observers, each observer did not begin processing until their average counts were less than one fish and median counts were within two fish of the training test key. Three observers reviewed the data, two were authors and the other a technician with the research team.

DIDSON analysis

To determine whether river entry timing of fish differed based on fish size and year, the mean migration date was compared

Table 1
A summary by species of the number of individuals sampled in each size class (small ≤ 30 cm, medium > 30 cm and < 50 cm, and large ≥ 50 cm) and average total length during electrofishing surveys in the Boardman River in April 2019, May 2017, May 2019, June 2019.

		2017			2019			
Species	Total	Small	Medium	Large	Small	Medium	Large	Mean Length (SE) - cm
Atlantic salmon (*Salmo salar*)	1	0	0	0	0	0	1	71.7 (NA)
bluegill sunfish (*Lepomis macrochirus*)	1	1	0	0	0	0	0	12.7 (NA)
brown trout (*Salmo trutta*)	14	1	5	0	1	6	1	35.4 (3.1)
cisco (*Coregonus artedi*)	3	0	0	0	3	0	0	18.6 (0.3)
coho salmon (*Oncorhynchus kisutch*)	1	1	0	0	0	0	0	10.2 (NA)
common carp (*Cyprinus carpio*)	25	0	0	2	0	0	23	70.1 (1.3)
common white sucker (*Catostomus commersonii*) *	270	6	49	2	0	117	19	43.5 (0.4)
gizzard shad (*Dorosoma cepedianum*)	7	0	3	0	0	4	0	45 (0.7)
golden redhorse (*Moxostoma erythrurum*)	5	0	0	1	0	0	4	60.4 (2.3)
lake trout (*Salvelinus namaycush*)	11	0	0	0	0	0	11	71.2 (2.8)
longnose sucker (*Catostomus catostomus*) *	330	0	19	0	0	95	5	44.4 (0.2)
Northern pike (*Esox lucius*)	9	0	1	0	0	1	7	64.1 (5.8)
rainbow trout (*Oncorhynchus mykiss*)	228	26	3	0	139	17	42	28.7 (1.3)
rock bass (*Ambloplites rupestris*)	22	22	0	0	0	0	0	16.4 (0.9)
round goby (*Neogobius melanostomus*)	6	6	0	0	0	0	0	8 (0.8)
smallmouth bass (*Micropterus dolomieu*)	111	92	4	0	1	13	1	16.9 (1.1)
walleye (*Sander vitreus*)	31	0	1	2	0	3	24	60 (1.6)

* denotes species in which total length was subsampled.

Table 2
A summary by species of the number of passive integrated transponder (PIT) tags implanted from July 2018-June 2019 and the mean, standard error (SE), median, and range of the lengths of fish implanted in centimeters. Small (≤30 cm), medium (>30 cm and < 50 cm), and large (≥50 cm) fishes were implanted.

Species	Total	Small	Medium	Large	Mean (SE)	Median	Range
brown trout (*Salmo trutta*)	20	2	16	2	41.5 (2.2)	41.2	24.6–65.9
common carp (*Cyprinus carpio*)	31	0	0	31	70.3 (1.1)	70	61.5–86.4
common white sucker (*Catostomus commersonii*)	58	0	49	9	43.4 (0.7)	42.5	32.7–56.4
gizzard shad (*Dorosoma cepedianum*)	3	0	3	0	44.7 (0.6)	45.3	43.5–45.4
golden redhorse (*Moxostoma erythrurum*)	4	0	0	4	59 (2.3)	59.7	53.0–63.5
lake trout (*Salvelinus namaycush*)	16	0	0	16	70.7 (2)	69.4	58.5–87.0
longnose sucker (*Catostomus catostomus*)	55	0	52	3	44.8 (0.5)	44.5	40.2–66.5
Northern pike (*Esox lucius*)	8	0	1	7	67.2 (5.2)	68.5	39.9–87.5
rainbow trout (*Oncorhynchus mykiss*)	48	7	21	19	48 (2.6)	42.2	20.5–89.0
rock bass (*Ambloplites rupestris*)	10	10	0	0	18 (0.7)	17.4	15.4–22.0
smallmouth bass (*Micropterus dolomieu*)	28	10	17	1	33.3 (1.7)	34.2	18.5–50.5
walleye (*Sander vitreus*)	24	0	2	21	61.2 (1.8)	60.4	45.7–76.1

between size classes and year using cumulative frequency distribution models and nonlinear regression. Fish entry patterns were presented in the context of cumulative frequency distribution models because they compare the mean and spread between fish size and year. The sigmoidal function $CF = 1/1 + e_0^{0/bD + X}$ was used to represent the cumulative frequency (CF) distribution models, where D represents day of the year, X_0 represents the mean day of entry, and b represents the relative spread in the day of entry, such that increases in b represent more variability in day of entry among individual fish of a given size class. This model was fit to our observed cumulative frequency distribution in which X_0 and b were allowed to vary by year and size class.

To evaluate environmental cues that predict the entry of fishes of a given size class (small, medium, or large) within the Boardman River, daily entry and environmental conditions were assessed using a hazard analysis. The hazard analysis, traditionally used to assess the probability of an individual's survivability based on daily environmental factors (Yoder et al., 2004), has been adapted to assess the probability associated with a fish movement event in various applications (e.g. Castro-Santos and Haro, 2003; Goerig et al., 2019; Knights et al., 2004; McCann et al., 2018; McMillan et al., 2014). Due to its semi-parametric nature, the hazard analysis is robust to departures from normal distribution and also allows for covariates whose values vary over the period of observation (i.e., time-dependent covariates; Harrell, 2015). Because fishes experienced different environmental conditions throughout the season, and different species may be influenced by different environmental cues, the hazard analysis was used to estimate the "cumulative hazard" for each fish to determine what environmental trigger(s) most influenced fish entry. In this case, we applied the hazard analysis with the assumptions that fish were observed during upstream migration, and all fish were available to migrate on the first day of the observational period. Each size class was evaluated separately and stratified by year using the following general hazard model: $h(t) = h_o(t)e^{(\delta_1 A(t) + \delta_2 B(t) + \cdots + \delta_n N(t))}$ where h is the hazard function for an individual at time t (in days from when the DIDSON was first deployed) according to a baseline hazard function (h_o). Annual stratification assumes a different baseline hazard within strata and evaluates covariate effect on baseline of each year respectively (Fox, 2008). The covariates ($A, B, \ldots, N$) used in our analyses were time of day (TOD), river temperature (RT), river stage (RS), and river discharge (D), all of which were considered time-dependent covariates. Additionally, we included 24-hour change (Δ) in covariates for river temperature (Δ RT), river stage (Δ RS), and discharge (Δ D). Cox (1972) proportional hazards models were constructed using the "survival" package in R (Therneau, 2015). The coefficients (δ_{1-n}) represent the degree to which each covariate affected fish movement and the exponentiated coefficients ($\exp(\delta_{1-n})$), also known as hazard ratios, give the effect size of covariates.

All possible models were reduced to a subset of only biologically relevant models such that no model consisted of both a covariate and the 24-hour Δ value. A correlation analysis showed that discharge and river stage were not highly correlated (r = 0.02); therefore, both were used as covariates in our models. A total of 53 models were considered in the analysis. Akaike's information criterion (AIC) was used to rank all models tested to determine which variables most influenced fish movement for each size class (Burnham and Anderson, 2004). Only models within 2 AICc units of the best-fit model (Δ AICc) were considered. After a set of best models was obtained, the proportionality of each model's covariates (an assumption of proportional hazard models) was confirmed using the Cox.zph function in the survival package (Therneau, 2015) and visual examination of the Schoenfeld residuals.

PIT tag analysis

Raw PIT tag detection data were limited to only include detection histories within the DIDSON viewing period. Data were filtered into individual detection events to reduce potential bias of a few individuals contributing disproportionately to the total detection number as a result of longer residence time at or near the antenna array. Detection events were created using a threshold value of 300 seconds (i.e., multiple detections within a five-minute period were synthesized into a single detection event). Mean and mode days of detection were estimated. Summary statistics and visual evaluation were used in comparison to 2019 DIDSON data to further elucidate species composition within each size class.

All analyses were performed in R (version 3.5.2). Nonlinear regression analyses were conducted using the nlsLM function within the minpack.lm (version 1.2–1) R package (Elzhov et al., 2016). Hazard analyses were run using the Coxph function within the survival (version 2.43–3) R package (Therneau, 2015). For all AIC analyses, the AICc and aictab functions within the AICcmodavg (version 2.1–1) R package were used (Mazerolle, 2017). Figures were created with the ggplot2 (version 3.0.0) R package (Wickham, 2016) with the exception of the map, which was created using ArcMap©.

Results

DIDSON

River temperatures ranged from 2.6 to 22.6 °C (2017), 2.8 to 24.5 °C (2018) and 3.0 to 21.5 °C (2019). River stage ranged from 17652.8 to 17717.5 cm (2017), 17676.3 to 17725.4 cm (2018), and 17688.1 to 17771.0 cm (2019) above sea level. Discharge ranged from 6.7 to 19.0 m^3/s (2017), 6.8 to 17.1 m^3/s (2018), and 7.9to

17.7 m^3/s (2019; Fig. 2). A total of 17,617 fish were observed in the DIDSON footage during analysis with 6597, 5150, and 5902 fish observed entering in 2017, 2018, and 2019 respectively. Of all observations, 1130 were classified as large fish, 11,679 were classified as medium fish, and 4840 were classified as small fish. Fish of all size classes were observed in the river throughout the entire DIDSON deployment (Fig. 3).

Estimates of mean day of entry for each fish size class was consistent in 2017, 2018, and 2019 with the peak number observed within a sub-sample DIDSON clip occurring 2–22 days from the estimated mean (Table 3). The estimated mean day of entry for large fish varied from May 2 to May 16 (Julian day 122–136) when river temperatures ranged from 10.6 to 14.4 °C, discharge ranged from 7.2 to 10.1 m^3/s, and river stage ranged from 17692 to 17720 cm (Table 3). The estimated mean day of entry for small fish varied from April 15 to May 9 (Julian day 105–129) when river temperatures ranged from 6.6 to 14.6 °C, discharge ranged from 10.1 to 11.9 m^3/s, and river stage ranged from 17680 to 17720 cm (Table 3). The estimated mean day of entry for medium fish varied from April 26 to May 9 (Julian day 116–129) when river temperatures ranged from 10.1 to 14.6 °C, discharge ranged from 10.1 to 11.9 m^3/s, and river stage ranged from 17680 to 17720 cm (Table 3). Proportionality testing of the top models for each size class revealed non-proportional hazards were present in the top models for large and small size classes preventing reliable interpretation, but the models of the medium size class met these assumptions (Electronic Supplementary Material (ESM) Tables S1, S2, S3).

The most parsimonious model of medium fish entry rates included the covariates TOD, RT, Δ D, and RS and no other model came within two AICc scores (Table 4). In this model medium fish were 21% (CI: 15–27%) more likely to enter the river at sunset and 25% (CI: 20–29%) less likely at midnight in comparison to midday (Table 4; ESM Figure S1). Entry rates of medium fish at sunrise in comparison to midday failed to demonstrate a significant effect when considered independently. Medium fish entered the river at an increased rate of 6% (CI: 3–10%) for every 1 °C increase in river temperature, and 4% (CI: 2–6%) for every 1 m^3/s increase in discharge from the day prior. The model also demonstrated that river stage had a small negative effect on the entry rate of medium fish with a 1% (CI: 1–2%) reduction for every unit increase in river stage (Table 4).

Physical sampling of fish

Due to the variety of species and life stages observed, there was overlap in size classes among the species observed during the electrofishing surveys (Table 1). The electrofishing surveys that occurred within the DIDSON sampling period found eleven species in the large fish class, eight species in the medium fish class, and nine species in the small fish class. The most abundant species observed within the large size class were rainbow trout, walleye,

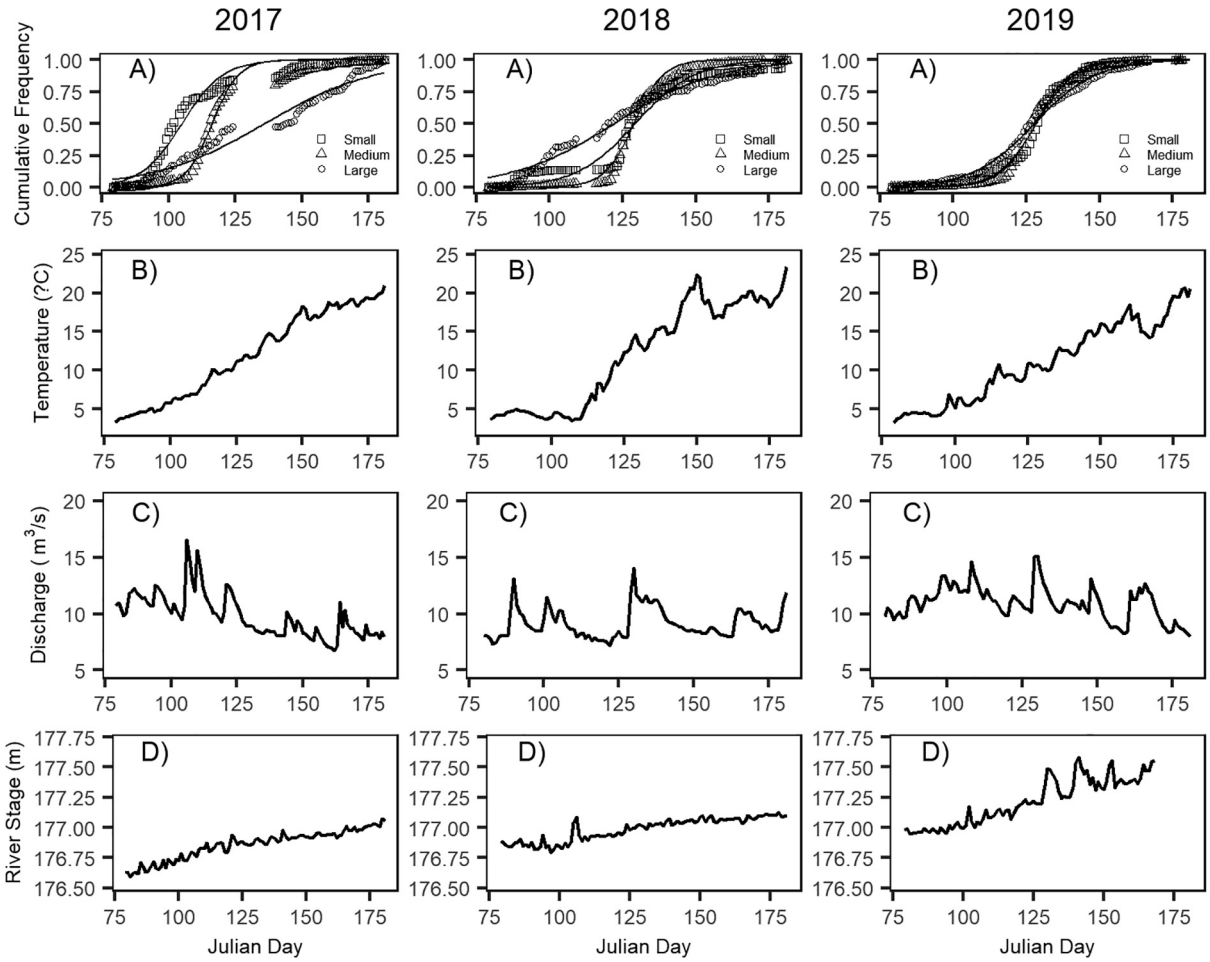

Fig. 2. Cumulative frequency plots of the proportion of fish observed on DIDSON in each size class (A), river temperature (B), discharge (C), and river stage (D) throughout the DIDSON deployment.

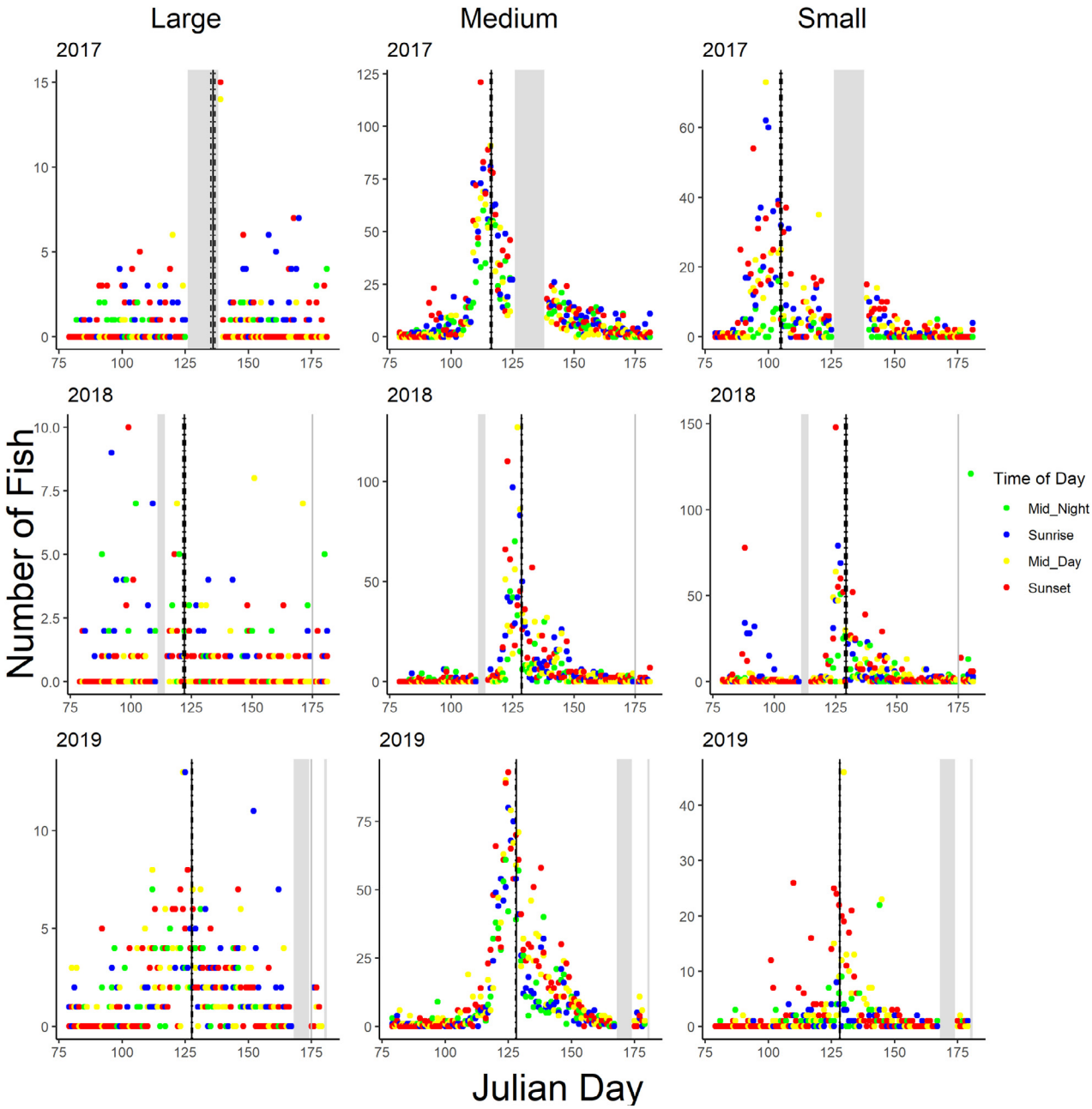

Fig. 3. Number of fishes in the large, medium, and small size classes observed in the Boardman River throughout the DIDSON deployment 2017–2019. Colors refer to the time of day when observations were made (i.e., mid-day, mid-night, sunrise, or sunset). Vertical bars indicate days in which DIDSON was not operational.

Table 3
A summary of time of day (TOD), river temperature (RT), discharge (D), and river stage (RS) associated with the Julian day (Peak) on which the highest number of fishes (# Fish) were enumerated in sub-sampled Dual-frequency IDentification Sonar clips and the cumulative frequency model parameters. Model values representing the mean (X_0), spread (b), and associated 95% confidence intervals of a model fit to the distribution of river entry timing of fishes in the Boardman River. Data summarized according to small (≤30 cm), medium (>30 cm and < 50 cm), and large (≥50 cm) fishes.

		Peak entry						*Mean entry*						
Year	Size class	Peak	# Fish	TOD	RT (°C)	D (m^3/s)	RS (cm)	X_0	CI (X_0)	b	CI (b)	RT (°C)	D (m^3/s)	RS (cm)
2017	Large	139	15	Sunset	14.4	8.2	17,688	136	135.1–136.8	20.8	20.0–21.5	14.4	8.5	17,692
2018	Large	99	10	Sunset	3.7	8.5	17,686	122	121.5–122.5	17.5	17.0–17.9	11.2	7.2	17,694
2019	Large	124	13	Midday	9.1	11.0	17,720	127.6	127.4–127.8	10.4	10.3–10.6	10.6	10.1	17,720
2017	Medium	112	121	Sunset	8.1	12.1	17,680	116.3	115.9–116.6	4.9	4.6–5.3	10.1	10.1	17,687
2018	Medium	127	127	Midday	12.7	12.4	17,700	128.7	128.5–129.0	5	4.8–5.2	14.6	11.9	17,702
2019	Medium	125	93	Midday	10.8	10.7	17,722	128.1	127.9–128.2	5.9	5.8–6.1	10.6	10.1	17,720
2017	Small	99	73	Midday	5.8	10.6	17,674	105	104.5–105.4	7.7	7.3–8.1	6.6	10.6	17,680
2018	Small	125	148	Sunset	12.4	14.1	17,698	129.2	128.6–129.7	10	9.5–10.5	14.6	11.9	17,702
2019	Small	130	46	Midday	10.0	14.7	17,749	128.4	128.2–128.5	6.5	6.4–6.7	10.6	10.1	17,720

Table 4
The most parsimonious models (Δ AICc ≤ 2) tested using an Akaike Information Criteria (AIC) approach on the effect of river temperature (RT), river stage (RS), discharge (D), time of day (TOD), as well as a 24-hour change (Δ) of the first three environmental variables (Δ RT, Δ RS, Δ D) on entry of medium fish (>30 cm and < 50 cm) into the Boardman River. K, Δ AIC_c, AIC_cWt, and N-event correspond to the number of covariates, corrected Akaike value, difference between each model's AICc and that of the best model, the model weight, and the number of events respectively within the model. The hazard ratios, lower and upper 95% confidence intervals (LL and UL, respectively), and *p* values correspond to covariates included within each fitted Cox Proportional Hazards model. A hazard ratio of 1 represents no effect on movement rates; values > 1 indicate greater entry rates; values < 1 indicate reduced movement rates.

Model	K	ΔAIC_c	AIC_cWt	N-event	Covariate	Coefficient	Hazard Ratio	LL (95%)	UL (95%)	p
TOD + RT + Δ D + RS	6	0	0.99	11,634	Δ D	0.04	1.04	1.02	1.06	0.000
					RS	−0.01	0.99	0.98	0.99	0.000
					RT	0.06	1.06	1.03	1.10	0.001
					TOD (Midnight)	−0.28	0.75	0.71	0.80	0.000
					TOD (Sunrise)	0.02	1.02	0.97	1.08	0.459
					TOD (Sunset)	0.19	1.21	1.15	1.27	0.000
Null	–	1061.4	0.00	0	–	–	–	–	–	–

common carp, white sucker, and lake trout. The most abundant species observed within the medium size class were white sucker, longnose sucker, rainbow trout, smallmouth bass, and brown trout. However, catch numbers for suckers were more than five times higher in comparison to any other species in the medium size class (Table 1). The most abundant species observed within the small size class were rainbow trout, smallmouth bass, and rock bass (Table 1).

PIT tag telemetry

A total of 305 fish were implanted with PIT tags in July 2018 - June 2019 (Table 2). The only species tagged that solely represented the small size class was rock bass (Table 2). The species tagged with an average total length (TL) in the medium size class consisted of brown trout, white sucker, rainbow trout, smallmouth bass, longnose sucker, and gizzard shad (Table 2). The species tagged with an average TL in the large size class consisted of common carp, northern pike, lake trout, walleye, and golden redhorse (Table 2).

The mean Julian day of detection for each species of PIT tagged fishes ranged from 123 to 172 (Table 5; Fig. 4). The mean day of PIT tag detection for longnose sucker (Julian Day 123; Table 5) and white sucker (Julian day 130; Table 5) on the PIT antennas align closely with the mean day of entry of medium fish (Julian Day 128; Table 3) estimated from the DIDSON. The peak day of entry for longnose and white sucker estimated from PIT tag detections (Table 5) coincided closely with the peak day of entry of medium fish estimated from the DIDSON analysis (Table 3). The only other species sampled in the medium size class (Table 1), with mean PIT tag detections within 10 days of the mean derived from DIDSON data, included brown trout, gizzard shad, and rainbow trout. However, the number of detections for these species was more than three times fewer than that of either sucker species.

Discussion

Our results show that it is possible to use DIDSON data and Cox regression to quantify the direction and magnitude of the effect that environmental covariates have on river entry for a subset of fish size classes defined *a priori*. However, the application of these models to DIDSON data is limited due to the inability to distinguish fish species. The evaluation of different size classes of fish was utilized with varying success to begin to deconstruct species composition. It is likely that our conclusive finding for medium fish resulted from the fact that this size group was primarily composed of white sucker and longnose sucker, which have similar life history traits and responses to environmental cues for migration.

Physical sampling of fish and data collected from the RFID system in 2019 shows corroborative evidence that the DIDSON analysis of medium fish was primarily detecting the spring sucker migration. The mean day of detection of these two species coincided more closely than any other species with the estimated mean entry of medium size class fish. Further, the physical collection of fish during the observational period showed drastically larger catch numbers for suckers in comparison to any other species in the medium size class (Table 1). This result supports other studies that show that although the migration of these species overlap, longnose sucker commonly migrate upstream earlier in the spring in comparison to white sucker (Geen et al., 1966; Klingler et al., 2003; Montgomery et al., 1983).

Our results demonstrate that across all years the covariates time of day, river temperature, river stage, and a 24-hour change in river discharge all had significant effects on river entry rates for longnose and white suckers. The most pronounced covariate effect on river entry (i.e., largest magnitude of effect) was associated with river temperatures, as each 1 °C increase in temperature corresponded with a 6% increase in entry rates of medium fish. It is well documented that longnose and white suckers migrate into rivers and

Table 5
Passive integrated transponder (PIT) tag detection event history by species giving the Julian day with the highest number of detections (Peak), number of detection event, the mean Julian day of detection, and the 95% confidence interval around the mean Julian day of detection in 2019. Detection events were created synthesizing multiple detections within a five-minute period into a single detection event.

Species	Peak	Detection events	Mean	95% CI
brown trout (*Salmo trutta*)	125, 126	4	127.8	123.3–132.4
common carp (*Cyprinus carpio*)	158	11	162.5	160.6–164.4
common white sucker (*Catostomus commersonii*)	123	37	129.9	128.2–131.5
gizzard shad (*Dorosoma cepedianum*)	132	2	129.0	123.0–135.0
longnose sucker (*Catostomus catostomus*)	121	28	122.5	121.7–123.2
Northern pike (*Esox lucius*)	152, 154	2	139.4	131.8–147.0
rainbow trout (*Oncorhynchus mykiss*)	99	8	127.4	123.9–131.0
rock bass (*Ambloplites rupestris*)	149, 157, 145	1	150.3	143.3–157.4
sea lamprey (*Petromyzon marinus*)	159	25	158.3	157.1–159.4
smallmouth bass (*Micropterus dolomieu*)	158	30	162.3	160.4–164.2
walleye (*Sander vitreus*)	174	3	171.8	168.3–175.2

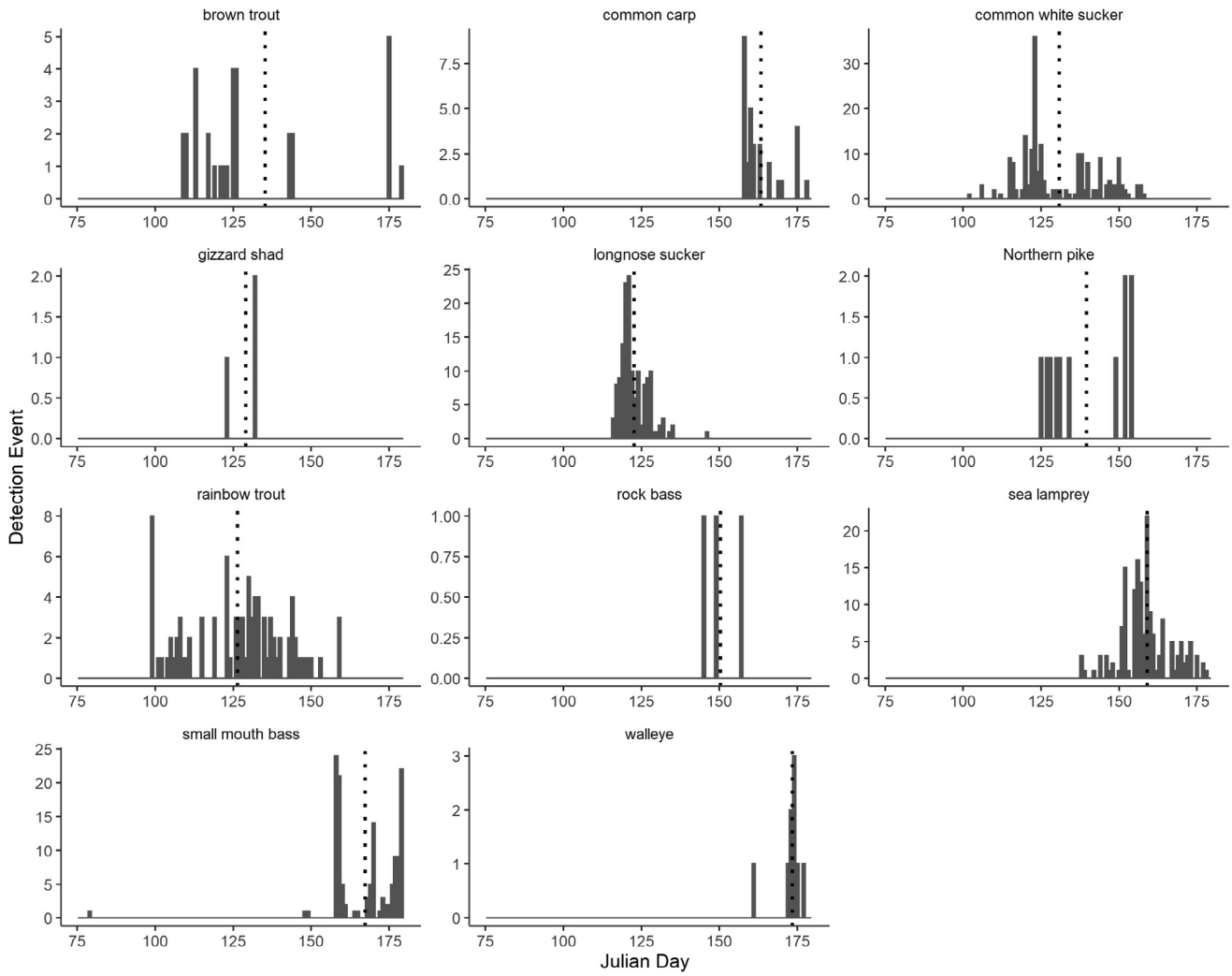

Fig. 4. Histogram of detection events by Julian day (78–181) of brown trout *Salmo trutta*, common carp *Cyprinus carpio*, common white sucker *Catostomus commersonii*, gizzard shad *Dorosoma cepedianum*, longnose sucker *Catostomus catostomus*, Northern pike *Esox lucius*, rainbow trout *Oncorhynchus mykiss*, rock bass *Ambloplites rupestris*, sea lamprey *Petromyzon marinus*, small mouth bass, and walleye tagged with passive integrated transponder (PIT) tags in the Boardman River. Dotted vertical line indicated the estimated mean day of detection. Detection events were created synthesizing multiple detections within a five-minute period into a single detection event.

streams in the spring for spawning across North America once a minimum threshold temperature range is reached and adequate water is present (Barton, 1980; Walton, 1980 as cited by Hamel et al., 1997; Corbett and Powles, 1983; Geen et al., 1966; Raney and Webster, 1942). The temperature threshold for white sucker varies across the literature between 7.2 °C (Raney and Webster, 1942) and 10 °C (Corbett and Powles, 1983; Geen et al., 1966). There are important thermal considerations engrained in the physiology of these animals. Egg incubation time is negatively related to temperature in suckers, and cold temperatures extend the duration of incubation often increasing egg mortality (Hamel et al. 1997). Stream temperatures of 10–13 °C were estimated to be a critical threshold at which spawning behavior (laying and fertilizing eggs) takes place for white sucker spawning in inland tributaries (Hamel et al., 1997). The peak days of entry observed for medium fish fell within the threshold to begin migration and spawning. The same is true for the estimated mean day of entry with the exception of 2018 which was within two degrees of observed optimal spawning temperature. This provides additional evidence that these fish are cueing in on an important environmental covariate related to their underlying physiology. We found no significant interaction between time and temperature during assumption testing, suggesting that perhaps once a critical temperature is reached increasing temperature increased the number of suckers entering the river. However, visual examination of the data (Fig. 3) shows that there is a clear peak, and the suckers eventually spawn and return to downstream lacustrine habitats. One possible explanation for this is that direction of travel was not recorded in the methods used in our study. While most movement observed was in an upstream direction, the DIDSON data likely are picking up both the upstream movement during entry and downstream movement as the suckers return to Lake Michigan near the end of the study period. Future studies should consider additional methods such as those presented in Magowan et al. (2012) that document the direction of travel of all individuals observed to further disentangle downstream and upstream migration to avoid this potential effect.

While the importance of temperature thresholds is well documented, Geen et al. (1966) found that both longnose and white sucker migrations were more closely associated with warming stream temperatures rather than water levels in an inland lake in British Columbia, Canada. Our top model demonstrates there is a multivariate relationship between both variables in addition to time of day and river stage. This result both corroborates Geen et al. (1966) and others through demonstrating the importance of temperature, but also shows that migration cues are multivariate rather

than univariate in nature. The resulting higher ranked model that considered the change in discharge over a 24-hour period rather than total discharge demonstrates that these fish are in fact cueing in on changes in discharge rather than some critical value of discharge.

The significant decrease in entry rates at midnight and increase in entry rates at sunset in contrast to midday also align with observations of Geen et al. (1966) who documented higher movement rates between noon and midnight with peaks occurring during the evening. Predation avoidance is one possible explanation for the significant diel patterns of migration. Other species of suckers have documented diel patterns attributed to mitigation of predation risk, in which greater movement rates have been observed during dark and crepuscular hours in comparison to daylight hours relating to forage and shelter habitats respectively (Booth et al., 2013). Although forage movements and spawning migrations have different underlying motivations, the perceived reduction to predation risk can be viewed as mutually applicable.

The significant negative effect of increasing river stage on entry rate is potentially the most difficult relationship to understand and warrants additional examination. This finding may relate more to region wide increase in water levels, both in Lake Michigan (Fig. 2) and across the Great Lakes during the study period (U.S. Army Corps of Engineers, 2020). As such, this finding likely represents a general effect of time within the observational period. The early arrival of fish in 2017 relative to the other two years shifts the collective distribution of arrival times left (i.e., earlier; Fig. 3). Fundamentally, as time progressed through the period of study each year, the fish enter the river and then leave at the completion of spawning. This relationship would likely be more negatively correlated but is likely mediated slightly by the weak positive correlation between river stage and discharge.

The inability of the most parsimonious models for small and large fish to meet the proportional assumptions of the Cox model shows that these covariate effects are not consistent across time within the model strata demonstrating the temporal variability in the response of fish entry to these covariates. It is logical that covariate effect may change across the analysis period as it encompasses a wide range of covariate conditions. For example, it may be reasonable to expect that initial increases in temperature may have a more profound effect on river entry early in the spring as temperatures approach a threshold value such as those well documented in suckers (Corbett and Powles, 1983; Geen et al., 1966; Raney and Webster, 1942) and those proposed for rainbow trout in Great Lakes tributaries (Workman et al., 2002). Further evaluation of the interaction effect with time confirms this effect in the small and large size classes. Exploratory analyses of additional strata within years failed to meet model assumptions while also constructing a meaningful model.

There are numerous potential reasons the large and small size classes failed to meet the proportionality assumption of the Cox models. First, species recognition was not possible with the DIDSON images, and, as a result, these size classes are composed of a mixed species assemblage. Different species are known to respond to environmental cues differently. The selection of the length range for each size class was intended to pick up the large sucker migration which our results demonstrate was successful. However, the greater variety of species in the composition of the small and large size classes likely introduced confounding variability in determining the roles these covariates play in inducing fish to enter a river system during the migratory period. Another possible confounding factor for the large fish is the timing of the analysis period. The large fish size class is primarily composed of rainbow trout with additional components of common carp, lake trout, white sucker and walleye. The reason for stream entry and migratory timing of each of these species varies widely (Workman et al., 2002; Bronte, 1993; Montgomery et al., 1983; Chizinski et al., 2016). The combination of this variability among species coupled with the fact that the primary migratory species in this group (rainbow trout) likely moved into the system before DIDSON deployment likely resulted in the variability observed when testing model assumptions.

Analysis of entry rates for small fish could have been further confounded by the release and presence of stocked fish. The MIDNR stocked an average combined total of 140,000 coho and Chinook salmon, and rainbow trout (mean TL = 151 mm) during every year of the study within the evaluation period (MIDNR, 2020). These stocking events likely had a confounding effect on the analysis of the small size class; however, the stocking dates did not coincide with the peak observations within any years suggesting that natural numbers of small fish are relatively higher than the number artificially planted.

Additionally, it's important to acknowledge some of the other implicit assumptions of the Cox regression models in the context of both a failure to meet proportionality assumptions in small and large size classes and the successful models for the medium size class. One such assumption in our model is that all fish were available to enter the river on the first day of analysis until that individual was observed (e.g., a fish that passed on day 180 was available on day 79). This assumption may only be true to a varying extent. All fish were undoubtedly present within the river, Lake Michigan, or another connected water body during the analysis period. As such, all individuals had the physical ability to move into the river and subsequent DIDSON field of view on the first day of analysis. However, the spatial extent of the pool of available individuals could add confounding effects. Spatially distant individuals cannot possibly cue in on localized covariates but are available to move towards the river system and subsequently be subjected physiologically to these covariates affects. The proportion of individuals available may increase through time due to spatial distribution contradicting the assumption that a diminishing proportion of individuals are available.

The use of DIDSON provides a valuable fisheries data acquisition method because it does not require any physical sampling of fish, fish behaviors are presumably not influenced due to lack of handling, and it provides the ability to capture data at a large temporal scale (See Table 4; Baumgartner et al., 2006). However, these perceived benefits may be outweighed in many situations by the limitations described here and current inefficiencies associated with processing the large volume of data to achieve this resolution across long periods of time. Additional investment in computer learning algorithms and automation have the potential to improve the efficiency at which these data can be processed but will continue to struggle with the confounding of species recognition because of the coarse resolution of the DIDSON imaging. Considering the current state of automated species recognition or lack thereof, future research on fine-scale timing and environmental drivers of river entry should consider the continued use of additional technologies such as telemetry when evaluating these metrics in a mixed assemblage of fusiform fishes. Some exceptions exist, as these confounding effects have been overcome to a varying extent when using DIDSON data on species with unique identifiable features such as locomotory mode (e.g., sea lamprey; McCann et al., 2018) and extreme differences in body size (e.g., sturgeon; Crossman et al., 2011) or when used in combination with other observational techniques (Frear, 2002) and multiple DIDSON units (Hughes, 2012). Our results demonstrate the variable utility of DIDSON data for identifying migration phenology and environmental triggers of river entry of a mixed assemblage of migratory fishes. The results of medium size fish are particularly promising for applying the predictive capability of the Cox regression in order to inform potential sorting processes associated with the future

FishPass and other fish passage efforts. However, the apparent confounding influences suggest the utility is restricted to recognizable species, systems with a lower diversity of migratory fish species, or the presence of migration sizes of relatively large magnitude.

A recent study on DIDSON automation demonstrated that advances in Echoview version 6 software (Echoview Software, Hobart, Tasmania) can partially automate fish identification and counting of DIDSON data with similar accuracy as manual inspection in a Great Lakes tributary (Eggleston et al., 2020). Previous to this work, the use of Echoview version 5.2 found automated methods were inaccurate compared to manual inspection, although a fairly consistent 2:1 relationship between manual inspection and automated methods was found (Petreman et al., 2014). The combined results of Petreman et al. (2014) and Eggleston et al. (2020) demonstrate both that DIDSON can be used to evaluate migratory timing and abundance in the Great Lakes and the progression of technical advancements continues to improve. However, to our knowledge automation of species identification with DIDSON remains limited to species of unique swimming mode and automation of fish measurements has not yet been achieved. These endeavors may never be achieved on the DIDSON platform due to its replacement by Adaptive Resolution Imaging Sonar (ARIS; Sound Metrics Corp., Bellevue, Washington, USA). Future research should strive to develop ways to automate measurement of length in association with automated fish count as the combination would lend itself to analysis of migratory species of specific size ranges as presented here.

Our results show the value of DIDSON data for elucidating general trends in migratory fish behavior. Passive data acquisition allowed for an increased understanding of the time that suckers, the largest component of the medium sized fish, migrate into the Boardman River and identified some of the associated environmental cues at relatively less effort compared to traditional fish sampling methods. For example, collecting similar trend data using a continuous trap or netting design would have required substantially more staff and effort. The trends in river entry timing identified and their relationship to environmental variables offer predictive guidance to the future operation of FishPass. This predictive guidance can be used to generally informing when sorting mechanisms will need to be prepared to handle large influxes of fishes. For example, the day of peak and mean entry of suckers varied only 13–15 days across all three years of study, suggesting that sorting and passage mechanisms need to be in place in advance of this two-week period. The relationship with the environmental variables identified further refines the general understanding of river entry timing and also allows for more precise predictive modelling of possible environmental scenarios. For example, years of warmer temperatures and earlier snow melt which result in higher river temperatures and earlier increases in river discharge may generally lead to earlier entry timing as suggested by the relationship found between those variables. This in turn would suggest that sorting may need to take place earlier. Additionally, the predictive capacity of the Cox proportional hazard model allows for various plausible environmental conditions to be modeled and more precise predictions to be made that can in turn be tested at FishPass.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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This manuscript is contribution 4 of FishPass. FishPass is the capstone to the 20-year restoration of the Boardman (Ottaway) River, Traverse City, Michigan. The mission of FishPass is to provide up- and down-stream passage of desirable fishes while simultaneously blocking or removing undesirable fishes, thereby addressing the connectivity conundrum. We are grateful to the primary project partners: Grand Traverse Band of Ottawa and Chippewa Indians, Michigan Department of Natural Resources; U.S. Army Corps of Engineers; U.S. Fish and Wildlife Service, U.S. Geological Survey, and Fisheries and Oceans Canada. We also extend sincerest thanks to the primary partner, the City of Traverse City. Without the city's support and the vision of the city commission, FishPass would not have been possible. Funding for this contribution came from the Great Lakes Restoration Initiative and the Great Lakes Fishery Commission. Thanks to Frank Dituri, James Garavaglia, and Brett Fessell of Grand Traverse Band of Ottawa and Chippewa Indians, Heather Hettinger of Michigan Department of Natural Resources, and Tyler Bruning of the U.S. Geological Survey for assistance with fish sampling and DIDSON data acquisition. Thanks to Kelly Boughner of the Great Lakes Fishery Commission for conducting manual inspection of 2019 DIDSON images. Thanks to Scott Miehls for providing a helpful review of the manuscript. Any use of trade, product, or firm names is for descriptive purposes only and does not imply endorsement by the U.S Government.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jglr.2021.04.003.

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